

Total No. of Questions :8]

SEAT No. :

P1443

[Total No. of Pages : 2

[6004]:714

B.E. (Artificial Intelligence and Machine Learning) (Honours)

SOFT COMPUTING AND DEEP LEARNING

(2019 Pattern) (Semester-VIII) (410303)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain how genetic algorithm are different from evolutionary programming? **[6]**

b) What are operators in genetic algorithm? List and explain GA operators in brief? **[6]**

c) Explain Encoding methods? **[6]**

OR

Q2) a) What are types of crossover and mutation techniques? **[6]**

b) Mention application area of genetic algorithm? **[6]**

c) What is difference between genetic algorithm and genetic programming? **[6]**

Q3) a) List out the strength and weakness of artificial neural network? **[6]**

b) Explain in brief architecture of multilayer feed-forward neural network? **[6]**

c) Explain how weights are updated in perceptron network? **[5]**

OR

Q4) a) Explain Types of Artificial Neural Network (ANN) **[6]**

b) Explain Single layer and multilayer Perceptrons. **[6]**

c) Explain Self-Organizing Map in neural network? **[5]**

P.T.O.

- Q5)** a) Explain why to use deep learning? [6]
b) Explain in brief architecture of deep network? [6]
c) Explain different Deep Learning libraries? [6]

OR

- Q6)** a) Explain Restricted Boltzman Machines in details? [6]
b) Explain different Deep Learning platform. [6]
c) Explain any one Use Cases of Deep Learning? [6]

- Q7)** a) Explain in detail Convolution Neural Network(CNN)? [6]
b) Explain different Properties of CNN representations? [6]
c) List and explain different Applications of CNN.? [5]

OR

- Q8)** a) Explain in details Recurrent Neural Network (RNN)? [6]
b) Explain Sequential processing LSTM model in details? [6]
c) Explain difference between CNN and RNN.? [5]

